

PE-EC802A — Mixed Signal Design

Semester VIII • 3L:0T:0P • 3 Credits

Module 4 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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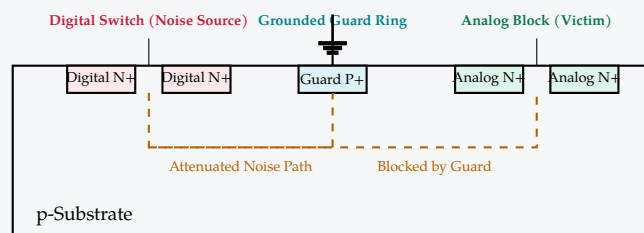
1. Mixed-Signal Layout Issues & Mitigation

In high-performance mixed-signal integrated circuits, fast-switching digital gates inject noise into the shared silicon substrate and power distribution networks, which severely degrades the resolution of sensitive analog blocks (like ADCs and low-noise amplifiers).

1.1 Substrate Coupling

Digital circuits consist of logic gates that switch states rapidly, drawing transient currents from the supplies. When the source/drain terminals of digital transistors transition, charge is coupled into the common p-substrate wafer via junction capacitances. This noise propagates through the conductive substrate to sensitive analog devices.

Cross-Sectional Model of Substrate Noise Coupling and Guard Ring Protection



Mitigation Strategy (Guard Rings): A guard ring is a low-impedance ring of diffusion surrounding the circuit blocks.

- **p+ Guard Ring:** Implemented in p-substrate, connected to a dedicated quiet analog ground (V_{SSA}). It collects hole currents flowing through the substrate from digital noise sources, shunting them to ground before they reach the analog block.
- **n+ Guard Ring in N-well:** Connected to quiet supply (V_{DDA}), collecting electron currents and isolating regions.

1.2 Power Supply Noise and Ground Bounce

Power and ground pads on an IC package exhibit parasitic inductance (L) and resistance (R). When digital gates switch simultaneously, they draw sharp current spikes (di/dt), causing a voltage fluctuation on the supply rails:

$$V_{\text{bounce}} = L \frac{di}{dt}$$

This fluctuation directly modulates the threshold voltages and bias conditions of analog circuits connected to the same supply rails.

Mitigation:

- **Split Supplies:** Separate supply pins and routing traces for analog (V_{DDA}, V_{SSA}) and digital (V_{DDD}, V_{SSD}) domains.
- **On-Chip Decoupling Capacitors (Decaps):** Placed near supply rails to act as local charge reservoirs, damping high-frequency voltage ripples.

1.3 3. Interconnect Parasitics & Signal Integrity

At high frequencies, on-chip and on-board wires cannot be treated as ideal conductors. They exhibit distributed resistance (r), inductance (l), and capacitance (c) per unit length.

Interconnect Lumped RLC Model

- **Resistance r (Ω/m):** From metal sheet resistance. Causes I^2R voltage drop and thermal loss.
- **Capacitance c (F/m):** From metal-to-substrate or metal-to-metal coupling. Acts as a low-pass filter, limiting signal bandwidth: $f_{-3\text{dB}} \approx 1/(2\pi r c l^2)$.
- **Inductance l (H/m):** Dominant at RF frequencies. Causes inductive ringing and reflections at impedance discontinuities.

Key Signal Integrity Issues:

- **Crosstalk:** Capacitive/inductive coupling between adjacent wires adds interference. Mitigated by inserting grounded shield traces between signal lines.
- **Reflections:** When a signal reaches an impedance discontinuity ($Z_L \neq Z_0$), part is reflected with coefficient:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Mitigated by matched termination ($Z_L = Z_0$, e.g., $50\ \Omega$ for coaxial cables).

- **Propagation Delay:** Signal propagation velocity is $v_p = 1/\sqrt{lc} \approx c_0/\sqrt{\epsilon_r}$. For FR4 PCB ($\epsilon_r \approx 4$): $v_p \approx 1.5 \times 10^8$ m/s.

2. Device Matching and Precision Layout

Matching of on-chip components (capacitors and transistors) is critical for accuracy in data converters and differential amplifiers.

2.1 1. Types of Mismatch

Mismatch is categorized as systematic or random.

- **Systematic Mismatch:** Caused by deterministic spatial gradients across the wafer (e.g., temperature variations, oxide thickness gradients, lithography light scattering).
- **Random Mismatch:** Caused by microscopic statistical variations (e.g., edge roughness, dopant atom fluctuations).

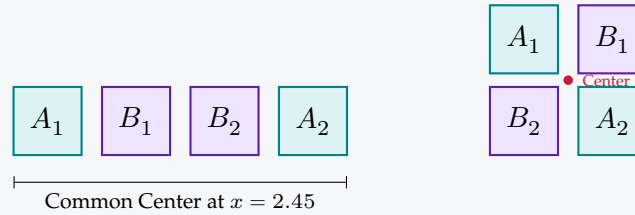
2.2 2. Common-Centroid Matching for Capacitors

To cancel the effects of linear spatial gradients (such as oxide thickness variations across a wafer), matching devices must be arranged symmetrically about a common central point (common-centroid configuration).

Consider matching two equal capacitors C_A and C_B . If they are placed side-by-side (A next to B), a linear gradient in oxide thickness will cause one to be larger than the other. If we split

each capacitor into two halves (A_1, A_2 and B_1, B_2) and arrange them in a one-dimensional or two-dimensional array, the gradient effects cancel.

Common-Centroid Capacitor Array Layout (ABBA)



By connecting $A = A_1 + A_2$ and $B = B_1 + B_2$, any linear variation in the horizontal or vertical direction affects both groups equally, preserving the ratio $C_A/C_B = 1$.

3. Low-Voltage Differential Signaling (LVDS)

3.1 Single-Ended vs. Differential Signaling

Single-ended signaling routes signals as a voltage relative to a common ground rail. Any ground bounce or external electromagnetic coupling adds directly to the signal, corrupting it.

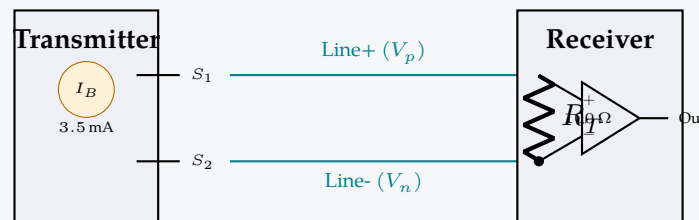
Differential signaling uses two complementary lines (V_p and V_n) to transmit information. The receiver detects only the difference $V_{diff} = V_p - V_n$.

- **Common-Mode Rejection:** Any noise coupled onto the board affects both lines equally. Since the receiver processes only the difference, this common-mode noise is rejected.
- **Reduced EMI:** The magnetic fields generated by the current in the two tightly-coupled differential traces are equal and opposite, cancelling out, which minimizes electromagnetic interference (EMI).

3.2 LVDS Transceiver Interface

LVDS is a high-speed, low-power physical layer interface standard. It operates by routing a constant current through a differential pair terminated by a resistor.

LVDS Transceiver Interface Schematic



Key Parameters:

1. **Current Loop:** The transmitter contains a 3.5 mA current source. Depending on the digital state, the switches route this current down either V_p or V_n .

- Termination:** The current flows through the transmission line and passes through the $100\ \Omega$ termination resistor R_T at the receiver, then returns via the other line.
- Voltage Swing:** The voltage swing across R_T is:

$$V_{diff} = I_{loop} \times R_T = 3.5\ \text{mA} \times 100\ \Omega = 350\ \text{mV}$$

This small voltage swing allows extremely fast switching and minimizes power consumption.

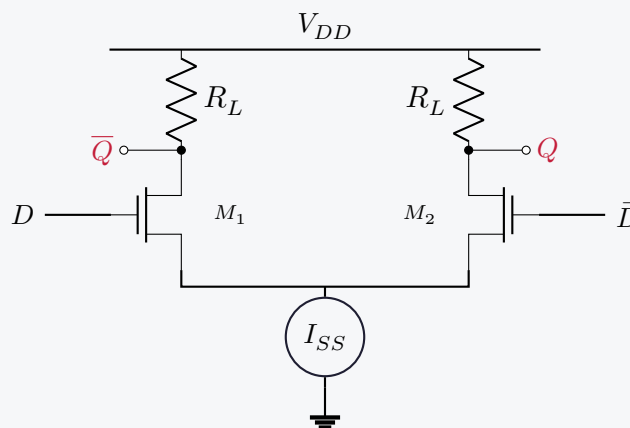
- Common-Mode Voltage:** The nominal common-mode voltage is set to $1.2\ \text{V}$, which is compatible with low-voltage supply rails.

4. Current-Mode Logic (CML) Signaling

Current-Mode Logic (CML) Concept

Current-Mode Logic (CML) routes a constant tail current I_{SS} through a differential transistor pair. The input data steers the current between two complementary output branches. Since the total current is always constant, the power rails see no transient spikes, and switching speed is limited only by the transistor f_T and the load $R_L C_{load}$ time constant — enabling multi-gigabit ($>10\ \text{Gbps}$) signaling.

CML Differential Pair Gate Schematic



Key CML Equations:

- Output Swing:** The differential voltage is set by the tail current and load resistance:

CML Output Swing and Bandwidth

$$\Delta V_{out} = I_{SS} \times R_L \quad (\text{Typical: } 200\ \text{mV} - 800\ \text{mV})$$

$$f_{-3\text{dB}} = \frac{1}{2\pi R_L C_{load}} \quad (\text{bandwidth set by RC time constant})$$

- Constant Current:** $P = V_{DD} \times I_{SS}$ is independent of switching activity — no transient supply spikes, making CML ideal for mixed-signal environments.

- 3. Common-Mode Level:** The output common mode is biased by the load: $V_{CM} = V_{DD} - I_{SS}R_L/2$, allowing direct DC cascading of multiple CML stages.
- **Applications:** SerDes transceivers, high-speed clock distribution, PCIe/Ethernet physical layers, multi-gigabit backplanes.

Quick Revision — Module 4

Layout & Signaling | PE-EC802A

Key Formulas at a Glance

Ground Bounce Voltage	$V_{bounce} = L \frac{di}{dt}$
LVDS Current Loop	$I_{loop} = 3.5 \text{ mA}$
LVDS Differential Swing	$V_{diff} = I_{loop} \times R_T = 350 \text{ mV}$
LVDS Termination Resistor	$R_T = 100 \Omega$
LVDS Common-Mode Voltage	$V_{CM} = 1.2 \text{ V}$

One-Line Definitions

- **Substrate Coupling:** Injection of digital switching noise into the shared silicon substrate, victimizing analog nodes.
- **Guard Ring:** A low-resistance diffusion ring surrounding sensitive analog components to collect substrate noise currents.
- **Ground Bounce:** Inductive supply noise caused by high di/dt transitions across package lead inductances.
- **Systematic Mismatch:** Mismatch on silicon caused by deterministic spatial gradients across the wafer.
- **Random Mismatch:** Mismatch on silicon caused by microscopic statistical variations (e.g., dopant fluctuations).
- **Common-Centroid Layout:** Symmetrical layout arrangement of matched devices around a single central point to cancel linear gradients.
- **LVDS:** A high-speed differential signaling standard transmitting data using a 3.5 mA current loop over a 100Ω termination.
- **CML:** Current-Mode Logic — routes a constant tail current I_{SS} through a differential pair for multi-gigabit signaling.
- **Interconnect Parasitics:** Distributed R, L, C of on-chip/board wires causing RC delay, reflections, and crosstalk.
- **Reflection Coefficient:** $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$; zero for matched termination ($Z_L = Z_0$).

High-Speed Signaling Interface Comparison

Parameter	CMOS (Single-Ended)	LVDS	CML
Signal Swing	V_{DD} (full rail)	350 mV diff	200–800 mV diff
Current Mode	Dynamic (CVf)	3.5 mA (const.)	I_{SS} (constant)
Max Speed	~1 Gbps	~1.2 Gbps	> 10 Gbps
Noise Immunity	Low (single-ended)	High (CMR)	High (CMR)
Power	Activity-dependent	Low-constant	Constant
Application	General IC logic	PCB data links	SerDes, backplane

Frequently Asked / Viva Questions

Q1. How does substrate coupling occur in mixed-signal ICs?

When digital circuits switch states, transient currents flow through junction capacitances into the shared p-type substrate. These currents travel through the silicon wafer and couple into the bodies of sensitive analog transistors, shifting their thresholds.

Q2. What is a guard ring, and how does it protect analog circuitry?

A guard ring is a ring of p+ diffusion (grounded) or n+ diffusion (connected to V_{DD}) surrounding a circuit. It collects substrate noise currents and shunts them to a quiet power supply before they can reach sensitive analog blocks.

Q3. Explain the physical mechanism behind ground bounce.

Ground bounce is caused by the parasitic inductance of package wires/pads. When multiple digital outputs switch states simultaneously, they draw high transient currents (di/dt). The inductive voltage drop ($V = Ldi/dt$) causes the ground potential to bounce.

Q4. How do split supply domains help in mixed-signal chips?

By separating the supply pads and lines for analog (V_{DDA} , V_{SSA}) and digital (V_{DDD} , V_{SSD}) domains, ground bounce and digital supply ripple are prevented from directly modulating analog circuit nodes.

Q5. What is the purpose of common-centroid layout in capacitor matching?

It arranges split capacitor cells symmetrically around a central point. Any linear spatial variation in oxide thickness or temperature affects both matching groups equally, cancelling the ratio error.

Q6. What is the difference between systematic and random device mismatch?

Systematic mismatch is caused by deterministic spatial gradients (like temperature or process variations) and can be minimized by layout techniques. Random mismatch is caused by random statistical variations (like dopant fluctuations) and can only be reduced by scaling up device dimensions.

Frequently Asked / Viva Questions (contd.)**Q7. Explain the working principle of LVDS signaling.**

LVDS uses a constant current source (3.5 mA) at the transmitter. The digital state controls switches to route this current through a differential trace pair. A $100\ \Omega$ resistor at the receiver terminates the lines, converting the current into a 350 mV voltage swing.

Q8. Why does differential signaling provide high noise immunity?

External noise couples equally onto both differential traces (common-mode noise). Since the receiver detects only the difference between the two lines, this common-mode noise is rejected.

Q9. State the nominal parameters of an LVDS interface.

Loop current: 3.5 mA; Termination resistance: $100\ \Omega$; Differential swing: 350 mV; Common-mode voltage: 1.2 V.

Q10. Why does LVDS generate very low Electromagnetic Interference (EMI)?

The currents in the differential pair are equal and opposite, flowing in close proximity. The magnetic fields they generate cancel each other out, minimizing electromagnetic radiation.